When Financial Struggles Lead to Borrowing: Understanding BNPL and Credit Risk

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GitHub Repository: https://github.com/NaniCode725/Buy-Now-Pay-Later-Credit-Risk-Prediction.git

Project Overview

Many people will agree that the cost of living in America has increased over the years. And Buy Now Pay Later services have become a very popular payment option, especially among consumers seeking flexibility in managing everyday expenses. While these services offer convenience, they have also raised concerns about their potential connection to financial hardship and increased credit risk. This project will examine the relationship between BNPL usage and indicators of financial wellbeing. The primary goal is to determine whether individuals who use BNPL services are already facing financial hardships. Through exploratory data analysis, data visualization, and predictive modeling, this study will help explain how financial circumstances influence the use of BNPL services.

Data Sources

List the datasets you will use. For each, include:

- **Dataset (one)- Buy Now Pay Later | Credit Risk Prediction**
- **Source: Consumer Credit Profiles**
- **Relevant fields - Age, Income, Credit Score, Debt- to- Income Ratio (DTI)**
- **Link - Buy Now Pay Later | Credit Risk Prediction**
- **Dataset (Two)- USA cost of living vs salary 2010-2024 per city**
- **Source: Longitudinal Regional Indexes**
- **Relevant fields- State, Year, Cost of Living Index, Income**
- **Link-** USA cost of living vs salary 2010-2024 per city

Research Objectives

- Are BNPL services utilized more by individuals already facing financial hardships?
- To what extent does BNPL usage reflect underlying financial stress and predict higher credit risk compared to traditional lending behavior?

These objectives are important because they explore whether individuals experiencing financial hardship are more likely to rely on Buy Now Pay Later services. The study seeks to demonstrate how some consumers are forced to shift debt from one obligation to another.

Data Preparation Approach

- The primary dataset contains Buy Now Pay Later (BNPL) credit risk records filtered for the USA. The secondary dataset tracks USA cost of living and average salaries over time.
- Both datasets will align to show the monthly income of individuals that use this service while also highlighting the debt that they already have. It will also give a picture of the monthly dept and cost of living.
- Once the datasets are fully prepared, I will conduct exploratory analysis to examine patterns in BNPL usage and financial risk, ensuring the final analytical file is clean, consistent, and ready for visualization and modeling.
- Outlines within individual variables such as monthly income and debt to income ratio will be identified using visualizations.
- To store and query my project data, established a rational schema consisting of two core tables derived from different data sources.

Table 1: bnpl users (Consumer Credit Profiles)

Stores demographic and transaction profiles for individual consumers.

Columns:

User id (Integer, Primary Key) , Age (Integer), Credit score (Integer), True monthly income (Float) , Location (String)

Table 2: state macroeconomics (Longitudinal Regional indexes)

Stores annual cost of living and localized wage data across the US.

Columns: state id (Integer, Primary Key), state(string), year (Integer), cost of living index (float), median salary net usd monthly(float)

Current Status

- Both required datasets have been successfully acquired and imported into the development environment
- Initial data exploration has been conducted to map out data types, columns schemas and null distributions. Also, duplicates have been removed.
- Developed an exploratory histogram analyzing the age distribution of BNPL users. Constructed a true monthly income histogram, this visualization highlights the economic concentration of consumers monthly income under $3000. Created a exploratory scatter plot evaluating data for 2024. This visualization mapped out low-income consumers by states.
- Evidence that the datasets can be successfully merged: Both datasets contain a clear geographic link: the primary dataset isolates transactions and consumers in the US (location == USA), while the secondary dataset breaks down macroeconomic data across individual US states (state)

Deliverables

- **Required Tasks:**
  Run the final aggregation on the salary and cost of living index numbers against net monthly wages. Complete and render cross-dataset visualizations.

Project Timeline

- Phase 1: Acquired and cleaned BNPL dataset
- Phase 2: Researched best visualizations to map the core consumer profile.
- Phase 3: Integrated second dataset containing US Salary and Cost of Living metrics. Filtered the modern economic indicators. Deeper analysis, refined visuals, draft report.
- Phase 4: Finalize deliverables, polish repo, pushed the finalized notebook updates to GitHub repo.

- The study was intentionally structured to evaluate localized consumer behaviors first before scaling out to a broader economic data.
- Geographic Exclusions and Rationales: Non-US locations were removed during the initial cleaning phase. This was necessary to maintain structural alignment with the secondary salary dataset, which included only US state-level data.